Contribution

Min:

For this lab, I restored the notebook since the original file was corrupted and could not run. Then I assigned roles to each member, including myself, to work on the lab. My roles in this lab were to set up the notebook, complete Task 1 (Confusion Matrix), and prepare the documents and notebook file for final submission. While running the notebook, I found that the line df['Age'].fillna(df['Age'].median(), inplace=True) no longer works in newer versions of pandas. I fixed it by replacing the column with df['Age'] = df['Age'].fillna(df['Age'].median()), and I learned that newer pandas uses Copy-on-Write, so an inplace method called on a selected column does not change the original DataFrame. For Task 1, I calculated the confusion matrix with confusion_matrix(y_test, y_pred) and visualized it as a seaborn heatmap. Across 179 test passengers, it showed 90 true negatives, 15 false positives, 20 false negatives, and 54 true positives. This taught me that an 80% accuracy score does not show what kind of mistakes a model makes: our model missed more real survivors than it wrongly predicted to survive, so it leans toward predicting Died. Finally, once everyone's sections were combined into one notebook, I renamed the file to the required naming convention, uploaded it to our GitHub repo, and submitted it.

Chris:

I contributed to Lab 07 by working on the model evaluation tasks, reviewing the confusion matrix and classification metrics, and examining how cross-validation measures model performance. I also worked on understanding the differences between precision and recall and why certain metrics are more important in specific real-world situations. Additionally, I completed the written responses by explaining the concepts in my own words. This work helped me strengthen my understanding of machine learning evaluation and how to interpret model results.

Camillie:

For this lab and Task 2, I contributed by generating a classification report using Python and the sklearn.metrics library to evaluate the Logistic Regression model. I worked through several coding errors, including indentation issues and undefined variables, before successfully running the code. Once the report was generated, I reviewed the precision, recall, F1-score, and accuracy to better understand the model's performance.

The model achieved an overall accuracy of 80%, which helped me understand how effectively it predicted passenger survival. This task strengthened my Python troubleshooting skills and gave me a better understanding of how classification models are evaluated.

Rigoberto:

For Lab 07, my individual technical contributions focused entirely on the implementation and analysis of Part 4: Cross-Validation. I wrote, debugged, and executed the 5-fold cross-validation routine using the `cross_val_score` function on the entire dataset. To satisfy the lab's documentation criteria, I added clear inline comments explaining how the data splits function, how the average performance is tracked, and how the standard deviation measures overall stability. On the theoretical side, I authored the analytical response for Knowledge Check Question 3, detailing how cross-validation eliminates the sampling bias and "sampling luck" inherent to static train-test splits. Finally, I handled group troubleshooting by diagnosing runtime environment state errors (`NameError`) and verifying that the entire notebook compiles in a correct, linear sequence for final submission.